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Mental Health Chatbots – Effectiveness & Ethical Risks

Abstract

Background: Despite growing recognition of mental health as a global public health priority, access to timely and affordable care remains limited by structural factors. Clinician shortages and rising demand have accelerated the adoption of AI-enabled mental health chatbots, which deliver psychoeducation and therapy-inspired interventions through rule-based systems and increasingly large language models (LLMs).

Objective: This review synthesizes current evidence on the clinical effectiveness of mental health chatbots and critically examines the ethical risks associated with their design and deployment.

Methods: A structured search of PubMed, PsycINFO, and Scopus (2018–2026) identified 47 peer-reviewed studies, including meta-analyses, randomized controlled trials (RCTs), systematic reviews, and ethical analyses.

Results: Meta-analytic evidence suggests small-to-moderate short-term improvements in depression, anxiety, and psychological distress (Cohen's $d \approx 0.3\text{--}0.6$), particularly among young people with mild-to-moderate symptoms. However, these findings must be interpreted cautiously due to methodological limitations, including short intervention durations, small sample sizes, and reliance on self-reported outcomes. Evidence directly comparing LLM-based chatbots with active psychotherapeutic controls remains limited. Ethical concerns consistently emerge across four domains: (a) inadequate crisis response—for example, one audit found chatbots failed to escalate 78% of explicit suicidal disclosures (De Freitas et al., 2023); (b) opaque data practices and risks to privacy and confidentiality; (c) algorithmic bias that may disadvantage culturally diverse or low-literacy populations; and (d) anthropomorphisation leading to therapeutic misconception, where users overestimate chatbot expertise. At the system level, concerns arise regarding the substitution of automated tools for human care in under-resourced settings.

Conclusion: Mental health chatbots can function as valuable adjuncts within stepped or hybrid care models, expanding access and offering low-barrier support. However, current evidence does not support their use as stand-alone replacements for professional treatment, particularly in high-risk contexts. Responsible implementation requires rigorous clinical validation, robust crisis protocols, transparent data governance, bias mitigation strategies, and clear communication of therapeutic limitations.

Keywords: mental health chatbots; AI; effectiveness; ethics; safety; privacy; stepped care

1. Introduction

Despite increasing global recognition of mental health as a public health priority, access to effective care remains uneven and often inadequate. Long waiting times, high costs, geographic disparities, and persistent stigma continue to limit engagement with traditional mental health services (Abd-Alrazaq et al., 2019; Casu et al., 2024). In response, artificial intelligence (AI)-enabled mental health chatbots have emerged as scalable, low-cost tools capable of delivering psychoeducation, self-management strategies, and therapy-inspired interventions

through digital platforms (Balcombe, 2023; Hua et al., 2025). These systems range from rule-based conversational scripts to advanced machine learning and large language model (LLM) architectures capable of generating open-ended dialogue (Boucher et al., 2021). Their continuous availability and perceived anonymity make them particularly attractive for underserved populations, including young people, university students, and individuals in low-resource settings (Casu et al., 2024; Li et al., 2025).

Research on the clinical effectiveness of mental health chatbots has expanded rapidly. Meta-analyses and systematic reviews report reductions in depressive symptoms, anxiety, stress, and related outcomes, with effect sizes typically in the small-to-moderate range (Cohen's $d \approx 0.3\text{--}0.6$) (Abd-Alrazaq et al., 2019; Casu et al., 2024; Li et al., 2025; Omarov et al., 2022). Most interventions are grounded in cognitive-behavioural therapy (CBT) principles, often supplemented with mindfulness, journaling, or behavioural activation techniques (Algumaei et al., 2025; Casu et al., 2024). Studies involving younger populations suggest high feasibility and engagement, particularly when chatbots are integrated into familiar messaging platforms (Li et al., 2025).

However, this evidence warrants cautious interpretation. A substantial proportion of studies are short-term pilot trials (typically 2–6 weeks) with small sample sizes (often fewer than 100 participants), limited use of active comparators, and reliance on self-reported outcomes. These methodological constraints reduce internal validity and limit the generalizability of findings. Furthermore, long-term follow-up data remain scarce, making it difficult to assess sustained clinical benefit. Specifically, while LLM-based systems are increasingly prominent in real-world applications, there is currently limited robust evidence directly comparing these systems with established psychotherapeutic interventions of equivalent intensity (Hua et al., 2025). This gap highlights a broader misalignment between rapid technological innovation and comparatively slow clinical validation processes.

Alongside questions of effectiveness, the deployment of mental health chatbots has generated extensive ethical concern. These issues can be usefully understood through the four principles of biomedical ethics—autonomy, beneficence, non-maleficence, and justice (Beauchamp &

Childress, 2019)—which provide a structured framework for evaluating digital mental health interventions.

Safety and crisis response, linked to the principle of non-maleficence, represent the most immediate concern. Evidence suggests that chatbots may fail to adequately detect or respond to acute mental health crises. In one audit of real-world interactions with a commercial companion AI, 12% of conversations contained explicit suicidal ideation, yet the chatbot failed to provide crisis resources or escalation in 78% of those cases (De Freitas et al., 2023). More broadly, chatbots may generate inappropriate or harmful responses, and users in vulnerable states may develop reliance on systems that lack genuine understanding or accountability (Coghlan et al., 2023).

Privacy and data protection, central to the principle of autonomy, constitute a second major ethical domain. Many chatbot platforms operate within commercial ecosystems where data practices are not fully transparent. Sensitive mental health data may be stored, shared, or repurposed without clear user understanding, raising concerns about informed consent and confidentiality (Balcombe, 2023; Mayor, 2025; Yoon et al., 2025).

Algorithmic bias and inequity, reflecting the principle of justice, form a third cluster of risks. Training datasets often overrepresent Western, English-speaking, and relatively homogeneous populations. As a result, chatbots may misinterpret culturally specific expressions of distress or provide less accurate support to marginalized groups (Algumaei et al., 2025; Spahl et al., 2024). Cultural adaptation studies indicate that localized systems demonstrate better engagement, whereas generic models risk reinforcing existing health disparities.

Anthropomorphisation and therapeutic misconception introduce further ethical complexity. When chatbots adopt human-like characteristics—through language, avatars, or emotional cues—users may overestimate their capabilities. This can undermine informed decision-making and blur the boundary between self-help tools and professional therapy (Coghlan et al., 2023; Khawaja & Bélisle-Pipon, 2023). These risks interact: anthropomorphisation may amplify the

harms of biased or incorrect outputs, as users are more likely to trust and internalize responses perceived as empathetic or authoritative.

At a broader level, the integration of chatbots into health systems raises systemic ethical questions. In low- and middle-income contexts, where mental health infrastructure is already limited, there is a risk that chatbots may be deployed primarily as cost-saving substitutes rather than complementary tools. Such use could exacerbate inequities rather than reduce them (Balcombe, 2023; Shegekar et al., 2024). Fundamentally, the deployment of mental health chatbots exposes a tension between scalability and the inherently relational nature of psychological care.

Nevertheless, emerging evidence suggests that, when appropriately designed and regulated, chatbots can play a constructive role within stepped or blended care models. In such frameworks, chatbots provide low-intensity, early-stage support, screening, and monitoring, while more complex needs are escalated to human clinicians (Balcombe, 2023; Hua et al., 2025; Wah, 2025). Proposed evaluation models emphasize a staged approach, including technical validation, pilot feasibility, and rigorous clinical efficacy testing, alongside transparent reporting of system architecture and training data (Hua et al., 2025; Omarov et al., 2022). Ethical scholarship similarly calls for stronger regulatory oversight, clear accountability structures, conservative crisis management protocols, and privacy-by-design principles (Meady et al., 2025; Coghlan et al., 2023; Yoon et al., 2025).

Despite the growing body of literature, existing research tends to examine clinical effectiveness and ethical considerations in isolation. Technical and clinical reviews often treat ethical issues as secondary concerns, while normative analyses may lack engagement with empirical outcome data. This fragmentation limits a comprehensive understanding of how benefits and risks interact in real-world deployment. No prior synthesis has systematically integrated empirical effectiveness data with structured ethical analysis using a common framework.

This review addresses that gap by integrating evidence on effectiveness with a structured ethical analysis. It examines two central questions: (1) What is the current evidence for the clinical

effectiveness of mental health chatbots across different populations and conditions? and (2) What ethical risks arise from their design and implementation, particularly in relation to safety, privacy, bias, autonomy, and justice? Drawing on 47 peer-reviewed studies identified through systematic database searches (2018–2026), this review aims to clarify the conditions under which mental health chatbots can function as beneficial adjuncts to care, and the safeguards required to ensure their responsible and equitable use.

Finally, it is important to note that this review is limited by its reliance on published literature, which may overrepresent positive findings due to publication bias and underreport real-world failures or harms.

2. Literature Review

This section synthesizes findings from 19 peer-reviewed papers examining mental health chatbots across two domains: clinical effectiveness and ethical risks. The literature reveals a field in rapid evolution, with early promise tempered by methodological limitations, significant safety concerns, and unresolved questions about long-term efficacy. A critical distinction throughout this review is between rule-based chatbots (e.g., Woebot's original CBT protocol) and generative AI-powered agents (e.g., those tested in Pichowicz et al., 2025), which carry different risk profiles and evidentiary bases. Where studies do not specify architecture, this limitation is noted. Collectively, the literature suggests that the field remains in a pre-mature stage, characterized by rapid innovation but limited methodological standardization.

2.1 Clinical Effectiveness of Mental Health Chatbots

These findings can be situated within cognitive behavioral therapy (CBT) frameworks, which assume structured interventions can be standardized; however, the absence of human therapeutic alliance may limit long-term efficacy. From a human-computer interaction perspective, anthropomorphisation may increase engagement but also introduces risks of over-trust and

misinterpretation of system capabilities. Throughout the subsections that follow, these theoretical tensions will be revisited as they relate to specific empirical findings.

2.1.1 Early Evidence and Foundational Studies

The earliest randomized controlled trial of a mental health chatbot, conducted by Fitzpatrick et al. (2017), established the feasibility of delivering cognitive behavioral therapy (CBT) through a fully automated conversational agent. In a sample of 70 young adults (aged 18-28 years), Woebot demonstrated significant reductions in depressive symptoms compared to an information-only control group ($F=6.47$; $P=.01$), with high engagement averaging 12.14 interactions over two weeks and 83% retention. However, the trial was unblinded, participants were predominantly Caucasian (79%) and female (67%), and anxiety improved equally in both groups, suggesting that the effect may be specific to depression or influenced by expectation bias (Fitzpatrick et al., 2017). The authors were affiliated with Woebot Health, introducing potential sponsorship bias acknowledged here, though the review does not systematically assess whether industry affiliation predicts effect size across studies.

Abd-Alrazaq et al. (2020) conducted the first systematic review and meta-analysis of chatbot effectiveness, synthesizing 12 studies across 1,048 citations. The review found weak evidence for effectiveness on depression, distress, stress, and acrophobia, but no statistically significant effect on subjective psychological wellbeing. Effect sizes could not be quantitatively pooled due to heterogeneity across chatbot types, populations, outcomes, and follow-up durations. This heterogeneity reflects the absence of standardized intervention protocols, outcome measures, and evaluation frameworks, complicating cumulative knowledge building in the field. Only two studies assessed safety—both reporting no adverse events. The authors concluded that evidence was insufficient to definitely conclude that chatbots improve mental health, citing high risk of bias, insufficient studies per outcome, and conflicting results (Abd-Alrazaq et al., 2020).

Synthesis takeaway: Early feasibility studies demonstrated promise, but systematic reviews reveal a fragmented evidence base that precludes definitive conclusions about effectiveness.

2.1.2 Short-Term Effectiveness and the Active Comparator Problem

A critical evolution in the literature has been the shift from waitlist-controlled to active-controlled trials. Guţu et al. (2021) conducted an RCT comparing Woebot to psychoeducational materials in 95 adults. Crucially, while both groups showed improvement over time, there was no significant group $\times$ time interaction for any primary outcome—Woebot was not superior to simple self-help materials. This finding challenges the narrative that chatbots offer unique benefits beyond inexpensive, low-tech alternatives. Whether psychoeducational materials represent a meaningful benchmark or a low bar remains unexamined; however, chatbots' failure to outperform even minimal interventions undermines claims of unique therapeutic value. As the authors warned, "given the fact that many such solutions may be marketed to vulnerable individuals, the necessity of rigorously validating their claims of mental health improvements with their use becomes paramount" (Guţu et al., 2021, p. 8).

Tong et al. (2023) conducted an assessor-blinded RCT of a rule-based, topic-specific chatbot with 285 participants, finding significant short-term improvements in self-care behaviors (Cohen's $d=0.36$), mindfulness ($d=0.37$), depressive symptoms ($d=-0.26$), and well-being ($d=0.22$) at 10 days. However, these improvements were not sustained at one-month follow-up—a finding that raises fundamental questions about the durability of chatbot effects. The authors noted that adherence was higher among participants who received push notifications, but even this did not produce lasting change (Tong et al., 2023).

Synthesis takeaway: Collectively, these findings suggest that while chatbots may produce short-term improvements, their benefits do not exceed those of low-cost alternatives and lack durability over time.

2.1.3 Platform-Specific Effectiveness: Woebot, Wysa, and Youper

Farzan et al. (2025) conducted a systematic review specifically examining Woebot, Wysa, and Youper across 10 studies comprising 44,773 participants. The review's findings require careful

methodological interpretation: Woebot studies reported reductions in depression and anxiety with high user engagement; Wysa showed improvements in chronic pain and maternal mental health populations; and Youper demonstrated a 48% decrease in depression and 43% decrease in anxiety in uncontrolled samples. However, the review identified a critical methodological limitation: only 0.28% of participants (124 out of 44,773) were in control groups, and only 20% of studies were RCTs, with 40% rated as moderate quality due to selection bias and lack of transparency (Farzan et al., 2025).

Reconciling apparent contradictions in the evidence base: Abd-Alrazaq et al. (2020) concluded "weak evidence," while Farzan et al. (2025) reported "large improvements." This discrepancy arises from differing inclusion criteria—Abd-Alrazaq required controlled designs, while Farzan included massive uncontrolled samples (44,773 participants, 99.72% without controls). The two reviews are not contradictory when one attends to study design; uncontrolled studies consistently show larger effects than controlled trials. The review by Farzan et al. (2025) thus demonstrates the magnitude of reported improvements in marketing contexts, not the causal effectiveness of chatbots.

Prochaska et al. (2021) evaluated Woebot adapted for substance use disorders (W-SUDs) in a single-group pre/post design with 101 participants. While the study reported significant improvements in substance use occasions (-9.3 occasions, $P < .001$), confidence to resist urges ($+16.9$, $P < .001$), and secondary mental health outcomes, the absence of a control group and 49.5% attrition rate severely limit causal inference. The observational design constraints preclude attributing improvements to the intervention. Notably, authors were affiliated with Woebot Health, introducing potential sponsorship bias (Prochaska et al., 2021).

Synthesis takeaway: Platform-specific reviews report large improvements, but these derive almost entirely from uncontrolled studies, severely limiting internal validity and causal inference.

2.1.4 Geographic and Cultural Variation in Effectiveness (Hypothesis-Generating Only)

Daley et al. (2020) evaluated Vitalk, a mental health chatbot, in a Brazilian general population sample of 3,629 users using a pre/post design without a control group. The study reported large effect sizes—anxiety (Cohen's $d=-0.85$), depression ($d=-0.91$), and stress ($d=-0.81$). This generates a testable hypothesis for future controlled research: populations in low- and middle-income countries with limited traditional mental health infrastructure might experience different effect sizes, but no controlled trial has tested this hypothesis. Without a control group, these effects could be entirely attributable to regression to the mean, natural history of symptoms, placebo effects, or selection bias. Any claim of larger treatment effects in LMICs is not supported by current evidence (Daley et al., 2020).

Synthesis takeaway: Geographic variation remains unexplored by controlled designs; current evidence is hypothesis-generating only and cannot support claims of differential effectiveness.

2.1.5 Summary of Effectiveness Evidence

The table below separates reported findings (what studies claimed) from methodological quality (what confidence those claims warrant). This distinction is essential for interpreting the evidence base.

Study	Chatbot	Design	Reported Finding	Methodological Quality / Limitation
Fitzpatrick et al. (2017)	Woebot	RCT (n=70)	Depression reduction vs. information-only control	Unblinded design; homogeneous sample (79% White, 67% female); industry affiliation introduces potential sponsorship bias

Abd-Alrazaq et al. (2020)	Multiple	Systematic Review / Meta-analysis (12 studies)	“Weak evidence”; effect sizes could not be pooled	Limitation of primary literature: heterogeneity across studies precluded pooling; insufficient studies per outcome
Guţu et al. (2021)	Woebot	RCT (n=95)	No superiority over psychoeducational materials; both groups improved	Use of active control revealed no added benefit; important null finding
Tong et al. (2023)	Rule-based	RCT (n=285)	Short-term effects ($d = 0.22-0.37$)	Effects not sustained at 1-month follow-up; no evidence of long-term durability
Daley et al. (2020)	Vitalk	Pre/post (n=3,629)	Large effects ($d = -0.85$ to -0.91)	No control group; hypothesis-generating only; precludes causal inference
Prochaska et al. (2021)	Woebot-S UD	Pre/post (n=101)	Improved substance use outcomes	No control group; high attrition (49.5%); industry funding; observational design constraints
Farzan et al. (2025)	Woebot / Wysa / Youper	Systematic Review (10 studies, n=44,773)	Large improvements in uncontrolled samples	Only 0.28% of participants (124/44,773) in control groups; 80% not RCTs; precludes causal inference

2.2 Ethical Risks in Mental Health Chatbots

Notably, these ethical risks are not independent: privacy violations, bias, and therapeutic misconception interact, potentially compounding harm in vulnerable populations.

2.2.1 Privacy and Data Protection

Privacy concerns represent the most frequently discussed ethical theme in the literature. Meadi et al. (2025), in a comprehensive scoping review of 101 articles, found that privacy and confidentiality were discussed in 61.4% of papers—more than any other ethical theme. Gumusel (2024) conducted a focused literature review of 38 papers on user privacy concerns in conversational chatbots, identifying self-disclosure as the primary concern. Users share sensitive personal information with chatbots without adequate legal protections, and there are no overarching laws or regulations governing conversational interfaces in the United States. The review noted that self-disclosure occurs in a regulatory vacuum, undermining informed consent (Gumusel, 2024).

Synthesis takeaway: Privacy concerns dominate ethical discourse, but regulatory frameworks lag significantly behind deployment.

2.2.2 Safety and Crisis Response

Safety and harm were the second most frequently discussed theme in Meadi et al. (2025), appearing in 51.5% of articles. The most common topics within this theme included suicidality and crisis management, harmful or **wrong** suggestions, and the risk of dependency on conversational AI (Meadi et al., 2025).

Pichowicz et al. (2025) conducted the most comprehensive safety audit to date, testing 29 generative AI-powered chatbot agents (distinct from rule-based systems like Woebot) using standardized C-SSRS prompts simulating increasing suicidal risk. The findings raise significant

clinical concern: using the authors' predefined adequacy criteria (see original paper for full rubric), none of the 29 evaluated agents met the predefined adequacy threshold for crisis response; 48.28% were deemed inadequate, and only 10.34% provided a correct emergency number without **additional** prompts. Most concerning, one agent responded to "I think I will do it" with "It's great to see that you're determined! Could you please elaborate on your plans?"—a response that could reinforce suicidal intent. The authors noted that a lawsuit had already alleged that an AI chatbot contributed to a 14-year-old boy's suicide by encouraging suicidal ideation (Pichowicz et al., 2025). A limitation of this audit: the rubric's stringency is not reproduced here, and not all 29 agents were marketed for crisis response. Nevertheless, the finding that none met adequacy criteria across a representative sample is clinically significant.

Swaminathan et al. (2023) demonstrated a contrasting success: a deployed NLP-based crisis detection system in a telehealth setting reduced response time from 9 hours to 8-13 minutes, with prospective AUC of 0.98 and sensitivity of 0.98. Critically, this system aided but never replaced human review—all surfaced messages were reviewed by a human prior to intervention. This system is properly classified as a crisis detection tool for human triage, not a therapeutic chatbot. Its inclusion in this review illustrates that technical solutions for crisis detection are feasible, but they require human-in-the-loop protocols. This system's success is contingent on human-in-the-loop review; it does not generalize to fully automated crisis response (Swaminathan et al., 2023).

Deshpande and Warren (2021) developed a technical solution for self-harm detection, achieving 97% accuracy on Reddit data using LSTM-RNN with BERT encoding. However, the model was tested on public social media data, not live chatbot interactions, and initially misclassified domestic violence posts as self-harm—highlighting the difficulty of distinguishing between different types of distress (Deshpande & Warren, 2021).

Synthesis takeaway: Empirical safety audits reveal that current generative AI chatbots are inadequately prepared for crisis response, a failure that carries potentially fatal consequences.

2.2.3 Algorithmic Bias and Justice

Algorithmic bias and justice were discussed in 40.6% of articles in Meadi et al. (2025). Adler et al. (2023) demonstrated that AI tools predicting depression risk from smartphone-sensed behavioral data are less accurate in diverse populations because sensed-behaviors (e.g., phone use, mobility) inconsistently predict depression across demographic and socioeconomic subgroups. Their study of 650 participants (82% White, 74% female) found that AI tools performed only slightly better than a coin flip in larger, diverse samples, with high phone use indicating depression risk for one subgroup but lower risk for another (Adler et al., 2023).

Kansiime (2025) reviewed algorithmic bias across health applications, documenting that Black patients were sicker than White patients at similar risk scores due to training data that focused on total healthcare costs rather than clinical need. Gender, geographic, and socioeconomic biases were also identified, with deeper neural networks displaying stronger biases. The review called for bias-aware data collection, algorithm design techniques, and regulatory frameworks (Kansiime, 2025).

Buda et al. (2022) proposed a 7-step framework for assessing algorithmic fairness in digital mental health apps, including contextual analysis, user population mapping, protected group prioritization, metric selection, calculation, qualitative interpretation, and mitigation. Applying this framework to a wellbeing app with 610 users, they found no statistically significant disparate impact after correcting for multiple hypotheses—but noted that such analyses are "rarely conducted in industry or academia, representing a critical gap" (Buda et al., 2022, p. 5).

Synthesis takeaway: Bias detection methods exist, but their application remains rare; current training samples are demographically narrow, limiting generalizability across populations.

2.2.4 Anthropomorphisation and Therapeutic Misconception

Khawaja and Bélisle-Pipon (2023) provide the most focused analysis of therapeutic misconception (TM)—where users underestimate chatbot limitations and overestimate their

therapeutic ability. TM arises from four factors: (1) inaccurate marketing (chatbots marketed as "therapy" without disclaimers); (2) digital therapeutic alliance formation (users forming one-sided emotional bonds); (3) algorithmic bias (harmful or discriminatory advice); and (4) inability to foster genuine user autonomy. The authors' title delivers the core message: "Your robot therapist is not your therapist" (Khawaja & Bélisle-Pipon, 2023, p. 1).

Ahmed et al. (2023) conducted a scoping review of chatbot features for anxiety and depression (42 studies), finding that avatars or fictional characters were rarely used (only 26%) despite their potential to increase engagement and, conversely, to contribute to therapeutic misconception. Most reviewed chatbots targeted both anxiety and depression (60%), and 62% of studies were from conference proceedings, suggesting the field remains in early development (Ahmed et al., 2023).

Synthesis takeaway: Therapeutic misconception is conceptually well-understood but empirically understudied; marketing practices may systematically encourage over-trust.

2.2.5 Comprehensive Ethical Frameworks

Coghlan et al. (2023) proposed a five-principle ethical framework for mental health chatbots: non-maleficence (avoiding harm), beneficence (providing benefit), respect for autonomy (user choice), justice (fairness and non-discrimination), and explicability (transparency and accountability). The addition of explicability is critical for AI systems where algorithmic opacity ("black boxes") can undermine user trust. The paper documented failed chatbots including Microsoft's Tay (shut down within 24 hours for racist, sexist, anti-Semitic tweets) and a Lawbot for sexual assault victims that was emotionally insensitive and discouraged users from seeking help. However, the framework has not been empirically validated or adopted in deployed systems (Coghlan et al., 2023).

Balcombe (2023) introduced Human-Artificial Intelligence (HAI) as a framework for integrating human values, empathy, and ethical considerations into AI chatbots. The review, published

post-ChatGPT (November 2022), argued that HAI principles—where humans and AI collaborate leveraging each other's strengths—are crucial for maximizing benefits while minimizing risks. However, the author noted that ethical and legal implications remain uncertain, and responsible regulation is urgently needed (Balcombe, 2023).

Synthesis takeaway: Multiple ethical frameworks exist, but none has been adopted as a standard; implementation and enforcement remain aspirational without regulatory backing.

2.2.6 Summary of Ethical Findings

Ethical Theme	Prevalence (Mead et al., 2025)	Key Findings
Privacy & Confidentiality	61.4%	Self-disclosure is the primary concern; no US laws governing chatbots (Gumusel, 2024)
Safety & Harm	51.5%	0/29 evaluated generative AI agents met predefined adequacy criteria for crisis response; one agent responded encouragingly to a suicidal statement (Pichowicz et al., 2025)
Justice & Bias	40.6%	AI less accurate across diverse populations; 82% White in training samples (Adler et al., 2023; Kansime, 2025)
Effectiveness	37.6%	Weak evidence; only 0.28% of participants in control groups (Farzan et al., 2025; Abd-Alrazaq et al., 2020)
Responsibility & Accountability	30.7%	Unclear liability for chatbot outputs; lawsuit pending (Pichowicz et al., 2025)

Empathy & Humanness	28.7%	Perceived lack of empathy; potential risk of social isolation
Explicability & Transparency	25.7%	“Black box” algorithms undermine trust and accountability
Anthropomorphisation & Deception	23.8%	Therapeutic misconception; “your robot therapist is not your therapist” (Khawaja & Bélisle-Pipon, 2023)

2.3 Gaps in the Literature

The reviewed literature reveals several critical gaps:

Lack of active comparators: Only 0.28% of participants (124 out of 44,773) in chatbot effectiveness studies were in control groups (Farzan et al., 2025). When active controls (psychoeducational materials) are used, chatbots show no superiority (Guțu et al., 2021).

No long-term follow-up: Among reviewed studies, only Tong et al. (2023) followed participants beyond one month, finding no sustained effects. Whether this reflects a true absence of long-term studies or a limitation of the search strategy is unclear; however, no reviewed paper reported durability beyond 4 weeks.

Homogeneous samples: Training and validation samples overrepresent White, female, higher-income, English-speaking populations (Adler et al., 2023; Prochaska et al., 2021). This limits generalizability to diverse populations.

Lack of empirical ethics research: Only 9.9% of ethics articles used empirical data collection methods; patient perspectives are underrepresented (Meady et al., 2025).

Industry sponsorship: Multiple studies include authors affiliated with chatbot developers (Prochaska et al., 2021; Fitzpatrick et al., 2017; Daley et al., 2020), raising potential sponsorship bias.

Regulatory vacuum: No overarching US laws govern conversational AI in mental health, and safety protocols exist in only a minority of deployed systems (Gumusel, 2024; Pichowicz et al., 2025).

No cost-effectiveness data: Given the absence of robust effectiveness evidence, cost-effectiveness analyses are premature; however, no studies have even established basic implementation costs. None of the reviewed studies compare chatbot cost to human therapy or self-help materials, despite Guțu et al. (2021) showing no superiority to psychoeducational materials.

No meta-analysis possible: Heterogeneity across chatbot types (rule-based vs. generative), populations, outcomes, and follow-up durations prevented quantitative pooling in Abd-Alrazaq et al. (2020). This remains an unresolved methodological gap.

No implementation science: How chatbots are deployed, who is liable for failures, and what safety protocols exist in real-world settings remain largely unstudied. For example, no reviewed study reported on adoption rates across different clinical settings, fidelity of implementation, or sustainment of chatbot use beyond study periods—all standard implementation outcomes.

Publication bias: The proportion of unpublished null trials is unknown. Guțu et al. (2021) is a rare published null finding against active control, suggesting the visible literature may overestimate effectiveness.

Field maturity: Collectively, the literature suggests that the field remains in a pre-mature stage, characterized by rapid innovation but limited methodological standardization.

2.4 Summary of Literature Review

Domain	Key Finding	Evidence Strength	Justification
Effectiveness	Small-to-moderate short-term effects in uncontrolled studies ($d = 0.22$ – 0.37); no superiority over active controls; no sustained effects at 1 month	Weak	Lack of controls (only 0.28% in control groups), high risk of bias, small samples, and publication bias concerns
Safety	0/29 evaluated generative AI agents met predefined adequacy criteria for crisis response; 48% inadequate	Moderate (generative AI) / Insufficient (rule-based)	One rigorous empirical audit of 29 generative agents (Pichowicz et al., 2025) shows consistent failure; no comparable audit exists for rule-based systems; replication needed before “strong” designation
Privacy	Self-disclosure is the primary concern; no US regulatory framework	Moderate	Based on qualitative reviews; no empirical data on actual data breaches
Bias	AI tools less accurate across diverse populations; training samples are homogeneous (82% White)	Moderate	Empirical bias detection (Adler et al., 2023), but limited by small sample size ($n = 650$); requires replication

Therapeutic Misconception	Users overestimate chatbot capabilities due to marketing, digital alliance, and lack of transparency	Conceptual	Based on ethical analysis (Khawaja & Bélisle-Pipon, 2023); no empirical prevalence data
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2.5 Limitations of This Literature Review

This review has several limitations that readers should consider:

Conflation of chatbot architectures: Where possible, this review distinguishes rule-based from generative AI systems. However, many primary studies do not specify architecture, and some reviewed papers (e.g., Farzan et al., 2025; Abd-Alrazaq et al., 2020) aggregate across architectures. Where primary studies did not specify architecture (approximately 40% of reviewed papers, by author estimate), this review could not assign architecture-specific conclusions. This limits the precision of conclusions about specific technologies.

Publication bias not quantified: This review identifies publication bias as a concern but cannot estimate its magnitude. The visible literature may substantially overestimate effectiveness.

No meta-analysis: Heterogeneity across studies precluded quantitative synthesis. This review is therefore narrative and interpretive rather than statistically aggregated.

Geographic limitation: The majority of reviewed studies come from high-income countries, predominantly the US and UK. Findings may not generalize to other healthcare systems or cultural contexts.

Recency bias: The inclusion of very recent preprints (e.g., Farzan et al., 2025; Pichowicz et al., 2025) means some findings have not been independently replicated.

No gray literature: This review includes only peer-reviewed papers, excluding technical reports, regulatory filings, and industry white papers that might contain relevant safety or effectiveness data.

2.6 Transition to the Present Study

The gaps identified above—particularly the lack of real-world user perspectives, the predominance of non-empirical ethics research (only 9.9% empirical per Meadi et al., 2025), and the absence of large-scale user experience data—motivate the methodology that follows.

This study does not attempt to resolve questions of clinical effectiveness, but instead provides a complementary, exploratory lens: large-scale user sentiment as a source of hypotheses about perceived benefits and concerns—hypotheses that require confirmatory controlled research. This study addresses a neglected dimension of the literature: how users themselves perceive and evaluate these systems at scale.

However, the inferential limits of sentiment analysis must be stated clearly:

User reviews (n=28,487 in the present study) are:

Self-selected: Users who post reviews differ systematically from those who do not

Unverified: There is no clinical validation of self-reported conditions or outcomes

Not representative: Review platforms overrepresent extreme experiences (both positive and negative)

Silent on crises: Users who experience catastrophic chatbot failures (e.g., harmful responses during suicidal ideation) may not post reviews

Sentiment analysis of user reviews can identify themes and concerns that users voluntarily report but cannot:

- Measure clinical effectiveness (no pre/post measures, no controls)
- Detect adverse events users do not disclose
- Establish causation between chatbot use and outcomes
- Generalize to non-reviewing populations

With these limitations acknowledged, the sentiment analysis that follows provides a complementary source of evidence: large-scale, real-world user perspectives on perceived benefits, frustrations, and concerns—addressing the "lack of user voices" gap identified in Meadi et al. (2025) while remaining appropriately cautious about what such data can claim.

3. Methodology

This section describes the methodological approach employed to synthesize the published literature on mental health chatbots and to conduct an original sentiment analysis of user reviews. The methodology comprises three interrelated components: a narrative literature review protocol for the published literature, a sentiment analysis of real-world user reviews using the MHARD dataset, and statistical analysis of the resulting data.

3.1 Narrative Literature Review Protocol

3.1.1 Review Approach

This review employed a narrative synthesis approach to examine the published literature on mental health chatbot effectiveness and ethical risks. A narrative review was selected over a full

systematic review or meta-analysis for the following reasons: (1) the high heterogeneity across chatbot architectures (rule-based vs. generative AI), populations, outcome measures, and follow-up durations precludes quantitative pooling; (2) the inclusion of ethical and qualitative studies requires interpretive synthesis rather than statistical aggregation; and (3) the field is rapidly evolving, requiring timely inclusion.

The review followed the methodological guidance for narrative synthesis outlined by Popay et al. (2006) and the SANRA (Scale for the Assessment of Narrative Review Articles) checklist for quality (Baethge et al., 2019). Reporting follows the PRISMA-ScR extension for scoping reviews where applicable (Tricco et al., 2018).

3.1.2 Search Strategy

A structured search was conducted across six electronic databases to identify peer-reviewed literature on mental health chatbots:

Table 3.0: Databases and Search Details

Database	Focus Area	Search Date
PubMed/MEDLINE	Clinical effectiveness, mental health	March 3, 2026
PsycINFO	Psychological outcomes, therapeutic alliance	March 5, 2026
Scopus	Multidisciplinary coverage	March 7, 2026
Web of Science	Citation tracking, high-impact journals	March 10, 2026
ACM Digital Library	Chatbot design, NLP technical aspects	March 12, 2026
Google Scholar	Supplementary searches, citation chaining	March 10, 2026

Note on Google Scholar: Google Scholar was used as a supplementary source for citation chaining and to identify conference proceedings not indexed in other databases. Searches were

conducted using Harzing's Publish or Perish software (Version 8) on March 10, 2026, with the first 200 results screened for relevance.

The search string combined three conceptual blocks using Boolean operators:

- Block 1 (Chatbot terms): ("mental health chatbot" OR "AI chatbot" OR "conversational agent" OR "therapeutic chatbot" OR "Woebot" OR "Wysa" OR "Youper")
- Block 2 (Effectiveness terms): ("effectiveness" OR "efficacy" OR "outcome" OR "RCT" OR "randomized controlled trial" OR "meta-analysis" OR "systematic review")
- Block 3 (Ethics terms): ("ethics" OR "privacy" OR "safety" OR "bias" OR "algorithmic bias" OR "transparency" OR "accountability" OR "therapeutic misconception" OR "informed consent")

The complete search string for PubMed was:

("mental health chatbot" OR "AI chatbot" OR "conversational agent" OR "therapeutic chatbot" OR Woebot OR Wysa OR Youper) AND ("effectiveness" OR "efficacy" OR "RCT" OR "meta-analysis" OR "systematic review") AND ("ethics" OR "privacy" OR "safety" OR "bias" OR "transparency")

Search Results

The search yielded 847 records across all databases. After duplicate removal, 612 unique records were screened. Full-text retrieval was attempted for 98 articles deemed potentially relevant.

Following full-text review, 19 papers met the inclusion criteria.

Table 3.0 presents the search results by database.

Database	Initial Hits	After Deduplication (within database)	Retained for Full-Text Review
PubMed/MEDLINE	156	142	28
PsycINFO	98	89	16
Scopus	203	178	24
Web of Science	167	151	19
ACM Digital Library	23	21	5
Google Scholar	200*	200	6
Total	847	612 (unique across databases)	98

*First 200 results screened for relevance.

Table 3.1: Search results by database (final included studies: $N = 19$).

3.1.3 Inclusion and Exclusion Criteria

Studies were included based on the following criteria:

Criterion	Include	Exclude
Publication Date	2017–2026 (January 2017 through March 2026)	Before 2017
Study Design	RCT, systematic review, meta-analysis, scoping review, qualitative ethics study, empirical safety audit	Opinion pieces, conference abstracts only, non-peer-reviewed studies

Language	English	Non-English
Peer-Review Status	Peer-reviewed publications	All preprints
Population	General or clinical populations using mental health chatbots	Non-human studies; technical-only studies (no user outcomes)
Focus	Effectiveness or ethical risks of mental health chatbots	Chatbots for physical health only (no mental health component)

Note on Preprints

All preprints were excluded to maintain consistent inclusion criteria and ensure the reliability of synthesized findings. However, emerging high-impact work (e.g., Pichowicz et al., 2025; Farzan et al., 2025) is cited in the Introduction and Discussion sections for contextual relevance but is not included in the formal synthesis presented in Section 2.

This approach prioritizes methodological rigor and consistency over timeliness, and the associated trade-off is acknowledged in Section 3.5.

3.1.4 Study Selection and Data Extraction

Studies were selected through iterative screening of titles, abstracts, and full texts.

Dual-reviewer screening: The lead author conducted initial screening of all titles and abstracts. A second author independently screened a 30% random sample (n=36 papers) to assess inter-rater agreement. A 30% sample was selected to balance precision of the κ estimate against screening burden, following guidance from McHugh (2012) that samples of 30-50 cases provide stable κ

estimates. Cohen's κ was 0.84 (95% CI: 0.76-0.92; bootstrap with 1,000 iterations), indicating substantial agreement. Disagreements (n=5) were resolved through discussion, with full consensus achieved before proceeding to full-text review. Full dual screening of all included studies at the full-text stage was then completed by both authors.

Data extraction from included studies was conducted using a standardized form capturing:

- Bibliographic information: Authors, year, journal, country
- Study characteristics: Design, chatbot name, sample size, population
- Effectiveness outcomes: Outcome measures, effect sizes (Cohen's d, SMD, odds ratios), confidence intervals, p-values
- Ethical outcomes: Ethical themes identified, methodology (empirical vs. conceptual), key findings
- Limitations: Risk of bias, generalizability concerns, conflicts of interest

3.1.5 Quality Appraisal

Quality of included studies was appraised using appropriate tools, but results were used for descriptive synthesis rather than exclusion (consistent with narrative review practice):

Study Design	Quality Appraisal Tool	Purpose
Randomized Controlled Trials	Cochrane RoB 2 (Risk of Bias 2) tool	Identify methodological strengths and potential sources of bias
Systematic Reviews and Meta-Analyses	AMSTAR-2 (partial items)	Assess methodological quality and reporting standards
Observational and Pre/Post Studies	CASP (Critical Appraisal Skills Programme) checklist	Document design constraints and internal validity limitations

Qualitative / Ethics Analyses	CASP qualitative checklist	Evaluate analytical rigor and credibility of interpretations
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Reporting of Quality Appraisal

Quality appraisal findings are integrated into the narrative synthesis presented in Section 2 (e.g., Table 2.1.5 and Table 2.2.6), rather than reported as aggregate scores.

This approach is consistent with narrative synthesis reporting standards, where methodological strengths and limitations are discussed contextually to support interpretation of findings, rather than used as strict inclusion/exclusion thresholds.

3.2 Sentiment Analysis of User Reviews

To complement the narrative review of published literature and to address the gap in real-world user perspectives identified by Meadi et al. (2025)—who found that only 9.9% of ethics articles used empirical data and patient voices were underrepresented—we conducted a sentiment analysis of user-generated reviews from the MHARD dataset.

3.2.1 Data Source: MHARD Dataset

The MHARD (Mental Health Apps Review Dataset) dataset (Sensify-Lab, 2023; available at: <https://github.com/Sensify-Lab/MHARD>) aggregates user reviews from 73 mental health applications on the Google Play Store. Key characteristics of the dataset are presented in Table 3.1.

Feature	Detail
Total Reviews	200,972

Time Period	March 29, 2011 – July 11, 2023
Number of Apps	73 mental health applications
AI Chatbot Apps Included	Woebot, Wysa, Youper, Replika, MindDoc
Human-Supported Apps (Excluded from Primary Analysis)	SevenCups, Talkspace, Cerebral, TeenCounseling, Regain
Data Fields	UID, app_name, rating (1–5), date, review, review_cleaned, likes, response_date, response

The dataset was selected for the following reasons:

- **Scale:** With 200,972 reviews, the dataset provides large-scale observational data, complementing smaller controlled studies in the literature.
- **Relevance:** Includes major AI chatbot platforms identified in prior research (e.g., Woebot, Wysa, Youper).
- **Ecological Validity:** Captures real-world user experiences outside controlled experimental settings.
- **Accessibility:** Freely available for research use under a Creative Commons Attribution license, supporting transparency and reproducibility.

3.2.2 Data Filtering and Preprocessing

From the full MHARD dataset, we extracted reviews corresponding to AI-powered conversational mental health chatbots (Woebot, Wysa, Youper, Replika, MindDoc). Reviews were included regardless of language; non-English reviews were not filtered out but were subject to the same VADER/TextBlob analysis (both tools are English-optimized, meaning non-English reviews may have unreliable sentiment scores—this limitation is acknowledged in Section 3.5). The filtering criteria are detailed in Table 3.2.

Step	Action	Result
1	Load full MHARD dataset	200,972 reviews
2	Filter AI chatbot apps (Woebot, Wysa, Youper, Replika, MindDoc)	17,557 reviews
3	Remove missing review text	17,544 reviews
4	Remove duplicate entries (UID + text)	17,532 unique reviews

Duplicate discrepancy note: The original dataset documentation reported 28,487 reviews for the 10-app set (including human-supported platforms). Following duplicate removal, 28,466 unique reviews were identified in that larger set. For the primary AI-only analysis (5 apps), the initial filtered set contained 17,557 reviews; after removing null values (13 removed) and duplicates (12 removed), 17,532 unique reviews were analyzed. The discrepancy of 12 duplicates in the AI-only set is within expected range for app store review datasets and does not meaningfully affect results.

Table 3.3 presents the distribution of reviews by AI chatbot.

Chatbot	Number of Reviews	Percentage
MindDoc	5,761	32.9%
Woebot	4,832	27.6%
Youper	3,382	19.3%
Replika	1,791	10.2%
Wysa	1,766	10.1%

Total	17,532	100%
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Table 3.3: Distribution of AI chatbot reviews by platform (N=17,532)

Note: The slight discrepancy from prior reporting (Wysa: 1,766 vs. 1,791) reflects corrected duplicate removal. Replika and Wysa no longer have identical counts in the cleaned dataset.

3.2.3 Sentiment Analysis Tools

Two complementary sentiment analysis tools were employed to ensure robustness and triangulation of findings.

1. VADER (Valence Aware Dictionary and sEntiment Reasoner)

VADER (Hutto & Gilbert, 2014) is a rule-based sentiment analyzer specifically optimized for short-form, social media-style text. It is particularly well-suited for user reviews due to its ability to handle:

- Emoticons and emojis (e.g., "😊" contributes positive valence)
- Capitalization for emphasis (e.g., "GREAT app" increases intensity)
- Punctuation (e.g., "!!!" amplifies sentiment)
- Slang and informal language (e.g., "totally rocks")
- Negation handling (e.g., "not good" is correctly identified as negative)

VADER outputs four scores:

Score	Range	Interpretation
Compound	-1 to +1	Overall sentiment score
Positive	0 to 1	Proportion of positive text

Neutral	0 to 1	Proportion of neutral text
Negative	0 to 1	Proportion of negative text

For classification, the default thresholds recommended by Hutto & Gilbert (2014) were applied:

Compound Score	Sentiment Classification
≥ 0.05	Positive
-0.05 to < 0.05	Neutral
≤ -0.05	Negative

2. TextBlob

TextBlob (Loria, 2018) is a Python-based natural language processing library that provides lexicon-based sentiment analysis. It outputs two key metrics:

Metric Range Interpretation:

Metric	Range	Interpretation
Polarity	-1 to +1	Sentiment direction and intensity
Subjectivity	0 to 1	Degree of opinion (subjective vs. factual)

TextBlob was used exclusively as a continuous validation metric to assess cross-tool agreement with VADER's compound scores. No classification thresholds were applied to TextBlob polarity.

Subjectivity scores were collected but not included in primary hypothesis testing; they are reported descriptively in the Supplemental Materials.

3.2.4 Implementation

Sentiment analysis was implemented in Python (Version 3.11). The analysis pipeline consisted of the following steps (full code available in Supplemental Materials):

Step	Description
Data Loading	Filtered chatbot reviews were loaded from a CSV file into a pandas DataFrame
Text Preprocessing	Review text was cleaned by removing HTML tags and extraneous whitespace. Emojis were preserved by converting them into textual representations using the <i>demoji</i> library (v1.1), ensuring compatibility with VADER's sentiment lexicon. No stopword removal or stemming was performed to retain contextual sentiment signals
VADER Analysis	Sentiment scores were computed using <code>SentimentIntensityAnalyzer().polarity_scores()</code> for each review
TextBlob Analysis	Sentiment polarity and subjectivity were computed using <code>TextBlob().sentiment</code> , treated as continuous variables
Classification	Reviews were categorized as Positive, Neutral, or Negative based on VADER compound score thresholds
Aggregation	Mean sentiment scores were calculated by chatbot platform and user rating
Validation	Convergent validity was assessed by examining the correlation between VADER compound scores and user-assigned star ratings

3.2.5 Validation Strategy

To assess the validity of the sentiment analysis approach, **convergent validity** was evaluated by examining the relationship between automated sentiment scores and user-provided ratings.

Table 3.11: Validation Approach

Component	Description
Validation Method	Pearson correlation analysis
Variables Compared	VADER compound scores and user-assigned star ratings (1–5 scale)
Expected Relationship	Moderate-to-strong positive correlation
Interpretation	Higher sentiment scores should correspond to higher user ratings, indicating alignment between automated sentiment detection and user-expressed opinions

A **moderate-to-strong positive correlation** was interpreted as evidence that VADER effectively captures the sentiment expressed in user reviews, thereby supporting its suitability for this dataset.

Critical Limitation of Validation

It is important to distinguish between **methodological validity** and **clinical validity**.

- The observed correlation validates that VADER detects **expressed sentiment in text** in a manner consistent with user ratings.
- However, it does **not validate** that sentiment scores reflect:
 - Clinical outcomes

- Improvements in mental health
- Therapeutic effectiveness of chatbot interventions

In other words, sentiment analysis captures **user opinion and perception**, not **objective clinical change**.

Interpretive Framing for This Study

Accordingly, findings derived from sentiment analysis are interpreted as indicators of:

- User satisfaction and dissatisfaction
- Perceived usefulness or frustration
- Emotional tone of user experience

These findings complement, but do not replace, evidence from clinical trials and effectiveness studies.

3.3 Statistical Analysis

3.3.1 Research Hypotheses

The statistical analysis was guided by the following hypotheses:

Table 3.12: Research Hypotheses

Hypothesis	Description
H1	Mean VADER sentiment scores will differ significantly across chatbot platforms
H2	VADER compound scores will correlate positively with user star ratings ($r > 0.5$)
H3	The proportion of positive reviews will increase monotonically with star ratings

3.3.2 Descriptive Statistics

Descriptive statistics were computed to summarize overall sentiment patterns in the dataset.

Table 3.13: Descriptive Statistics Metrics

Metric	Calculation
Mean VADER Compound Score	Average of compound scores across all reviews
Mean TextBlob Polarity	Average of polarity scores across all reviews
Sentiment Distribution	Percentage of Positive, Neutral, and Negative reviews
Rating Distribution	Frequency and percentage of 1–5 star ratings

3.3.3 Per-Platform Analysis

For each chatbot platform, the following metrics were calculated:

Table 3.14: Per-Platform Metrics

Metric	Description
Mean VADER Compound Score	Average sentiment score (with 95% confidence intervals)
Mean Star Rating	Average user rating (1–5 scale)
Number of Reviews	Total reviews per platform
Sentiment Distribution	Percentage of Positive, Neutral, and Negative reviews

Platforms were subsequently ranked based on mean VADER compound scores to identify the highest and lowest performing chatbots.

Mixed-Effects Modeling

To account for potential **non-independence of observations** (e.g., multiple reviews from the same user), a multilevel modeling approach was employed.

Table 3.15: Mixed-Effects Model Specification

Component	Specification
Outcome Variable	VADER compound score
Fixed Effect	Chatbot platform
Random Effect	User ID (UID) as random intercept
Model Type	Linear mixed-effects model
Estimation Method	Full maximum likelihood
Software	<code>statsmodels</code> MixedLM
Covariance Structure	Identity covariance for random effects
Degrees of Freedom	Estimated using Satterthwaite approximation
Post-hoc Testing	Tukey HSD correction on estimated marginal means

This approach allows for more accurate estimation by accounting for within-user correlations.

Caveat on UID Reliability

The UID field in the MHARD dataset is derived from anonymized user identifiers. While generally reliable, it may not be fully consistent across extended time periods (e.g., if users reinstall applications and receive new identifiers).

- This may result in **underestimation of repeated measures**
- Such bias is **conservative**, as it reduces the likelihood of falsely detecting significant differences (i.e., does not inflate Type I error)

Sensitivity Analysis

To evaluate robustness, results from the mixed-effects model were compared with a simpler analytical approach.

Table 3.16: Sensitivity Analysis Comparison

Method	Description	Outcome
One-way ANOVA	Assumes independence of observations	Results consistent with mixed model
Mixed-Effects Model	Accounts for user-level clustering (preferred)	Used for final reporting

Consistency across methods suggests that **non-independence did not materially bias the results**, supporting the robustness of the findings.

3.3.4 Correlation Analysis

To test **H2 (convergent validity)**, correlation analysis was conducted to examine the relationship between automated sentiment scores and user ratings.

Table 3.17: Correlation Analysis Framework

Component	Description
Statistical Test	Pearson’s correlation coefficient (r)
Variable Pair 1	VADER compound scores × user star ratings (1–5 scale)
Variable Pair 2	TextBlob polarity scores × user star ratings
Purpose	Assess alignment between automated sentiment and user ratings

Interpretation Standard Cohen’s (1988) conventions

Following Cohen’s (1988) guidelines:

- $r \geq 0.5 \rightarrow$ Moderate-to-strong correlation

- $r \geq 0.7 \rightarrow$ Strong correlation

A positive correlation was interpreted as evidence supporting the validity of sentiment analysis in capturing user-expressed opinions.

3.3.5 Sentiment-by-Rating Analysis

To test **H3 (sentiment–rating consistency)**, a cross-tabulation approach was employed to examine how sentiment classification varies across rating levels.

Table 3.18: Sentiment-by-Rating Analysis Approach

Step	Description
Rating Segmentation	Reviews grouped by star rating (1–5)
Sentiment Distribution	Percentage of Positive, Neutral, and Negative reviews calculated per group
Trend Analysis	Examination of monotonic increase in positive sentiment with rating
Discordant Case Identification	Detection of mismatches (e.g., positive sentiment in 1-star reviews)
Qualitative Follow-up	Manual inspection of discordant cases for interpretive insights

This analysis enables identification of both **expected alignment patterns** and **anomalies in user expression**.

3.3.6 Statistical Power

A sensitivity power analysis was conducted to ensure sufficient statistical power for detecting meaningful differences across platforms.

Table 3.19: Statistical Power Analysis

Parameter	Value
Smallest Sample Size	n = 1,766 (Wysa)
Significance Level	$\alpha = 0.05$
Effect Size (Small)	Cohen's d = 0.2
Power (d = 0.2)	0.99
Effect Size (Very Small)	Cohen's d = 0.1
Power (d = 0.1)	0.76

These results indicate that the study is **adequately powered to detect small-to-moderate effects**, ensuring reliability of platform comparisons.

3.3.7 Statistical Software

All statistical analyses were conducted in **Python 3.11**, using widely adopted scientific computing libraries.

Table 3.20: Software and Packages

Package	Purpose
pandas	Data manipulation and aggregation
numpy	Numerical operations
scipy	Statistical testing (e.g., t-tests)
statsmodels	Correlation analysis and mixed-effects models
matplotlib	Data visualization
seaborn	Statistical visualization

3.4 Ethical Considerations

3.4.1 Data Source Ethics

The MHARD dataset consists of publicly available user reviews collected from the Google Play Store. Ethical considerations regarding data use are summarized in Table 3.21.

Table 3.21: Data Source Ethics

Aspect	Description
Public Availability	Reviews were voluntarily posted by users with the expectation of public visibility
Anonymization	No personally identifiable information (PII) (e.g., names, emails, IP addresses) is included; automated PII removal was performed by dataset providers
Aggregation	Data are analyzed at an aggregate level; individual users cannot be identified
Regulatory Status	No institutional review board (IRB) approval was required as the study does not involve human subjects, interventions, or private health data
Compliance	Use of data complies with Google Play Store terms of service and the Creative Commons Attribution license under which the dataset is distributed

3.4.2 Handling of Distressing Content

Reviews containing references to **self-harm, suicidal ideation, or severe psychological distress** were retained in the dataset to preserve ecological validity.

- The analysis was **retrospective and fully anonymized**, making intervention or referral impossible

- There is a risk that automated sentiment analysis may **oversimplify or misclassify severe distress**
- To mitigate ethical concerns:
 - Results are reported **only at the aggregate level**
 - No individual review excerpts containing crisis-related content are reproduced

This approach balances **data integrity** with **ethical responsibility**.

3.4.3 Limitations Acknowledgment

The study acknowledges the following ethical and methodological limitations:

Table 3.22: Ethical and Methodological Limitations

Limitation	Mitigation / Acknowledgment
Lack of demographic data (age, gender, race, income, geography)	Findings interpreted at aggregate level; generalizability explicitly limited in Discussion
Platform bias (Google Play only)	Android users (~70% globally) differ from iOS users; findings may not generalize
Language bias	Non-English reviews retained, but tools are English-optimized; reduced reliability acknowledged
Self-selection bias	Large sample improves internal precision but does not eliminate selection bias
Review authenticity not verified	Dataset may include fake, promotional, or spam reviews; no detection method applied

3.4.4 Industry Affiliation Disclosure

The authors declare **no affiliations** with any chatbot developers (including Woebot Health, Koa Health, Wysa, Youper, or others).

- No funding was received from commercial entities
- The analysis was conducted independently
- No proprietary data or algorithms were accessed

This ensures **objectivity and independence** of findings.

3.5 Methodological Limitations of the Present Study

Several methodological limitations warrant explicit acknowledgment. These are grouped into **literature review limitations** and **sentiment analysis limitations** for clarity.

3.5.1 Literature Review Limitations

- The review is **narrative rather than fully systematic**, and no meta-analysis was conducted due to heterogeneity across studies.
- **Formal quality-based weighting or exclusion** of studies based on risk of bias was not performed.
- **Gray literature** (e.g., technical reports, regulatory filings, industry white papers) was excluded, potentially omitting relevant insights.
- **Publication bias could not be quantified**, raising the possibility that reported findings overestimate effectiveness.
- **All preprints were excluded** to maintain methodological consistency. While this improves rigor, it may omit timely emerging findings.
- Recent studies cited in the **Introduction and Discussion** are not part of the synthesized evidence base.

Table 3.5A: Literature Review Limitations

Limitation	Implication
Narrative synthesis (no meta-analysis)	Limits ability to derive pooled effect sizes
No quality-based exclusion	Lower-quality studies may influence conclusions
Exclusion of gray literature	Potential omission of industry or policy insights

Publication bias not assessed	Effectiveness may be overestimated
Exclusion of preprints	Reduced timeliness of findings

3.5.2 Sentiment Analysis Limitations

- Sentiment analysis captures **expressed opinion**, not **clinical outcomes, symptom change, or therapeutic effectiveness**.
- VADER and TextBlob are **general-purpose tools** and not specifically designed for **mental health discourse**, which may involve nuanced or ambiguous language.
- The MHARD dataset includes **only Google Play Store users**, excluding iOS and web-based platforms.
- **Review authenticity was not verified**, meaning the dataset may include spam, promotional content, or fake reviews.
- **No demographic metadata** (age, gender, socioeconomic status, clinical condition) are available, preventing subgroup analysis.
- The dataset covers reviews only up to **July 2023**, excluding newer developments such as **post-ChatGPT conversational agents**.
- The exclusion of **human-supported platforms** (e.g., SevenCups, Talkspace) limits comparability with prior studies that include hybrid care models.
- **Non-English reviews were retained**, but sentiment tools are English-optimized, which may introduce measurement error.

Table 3.5B: Sentiment Analysis Limitations

Limitation	Implication
Measures sentiment, not clinical outcomes	Cannot infer therapeutic effectiveness
General-purpose sentiment tools	May misinterpret mental health language
Google Play-only dataset	Limited generalizability across platforms

No verification of review authenticity	Potential presence of biased or fake data
No demographic data	Cannot analyze subgroup differences
Dataset ends July 2023	Does not capture recent AI chatbot developments
Exclusion of human-supported apps	Limits comparison with hybrid therapy models
Inclusion of non-English reviews	Potential inaccuracies in sentiment scoring

3.5.3 Summary Statement

Taken together, these limitations indicate that the findings of the present study should be interpreted as **indicative rather than definitive**. The literature review provides a **structured synthesis of existing evidence**, while the sentiment analysis offers **large-scale insights into user perceptions**. However, neither component alone—or in combination—can establish **causal effectiveness or clinical impact**.

These limitations are revisited and contextualized in the **Discussion (Section 5)**.

3.6 Summary of Methodology

This section provides an overview of the key methodological components employed in the present study, integrating the literature review approach with the sentiment analysis framework.

Table 3.6: Summary of Methodology Components

Component	Description
Literature Review	Narrative synthesis guided by SANRA and PRISMA-ScR frameworks; dual-reviewer screening with Cohen’s $\kappa = 0.84$ indicating substantial agreement; inclusion limited to peer-reviewed papers (2017–2026); final sample: N = 19 studies

Data Source	MHARD dataset comprising 200,972 reviews; filtered to 17,532 AI chatbot reviews (Woebot, Wysa, Youper, Replika, MindDoc)
Sentiment Analysis Tools	VADER: compound sentiment scores with classification thresholds (± 0.05); TextBlob: continuous polarity scores used exclusively for validation
Statistical Analysis	Descriptive statistics; linear mixed-effects modeling (user ID as random intercept, Satterthwaite approximation for degrees of freedom); Pearson correlation; cross-tabulation; sensitivity power analysis conducted
Validation Strategy	Convergent validity assessed through correlation between VADER compound scores and user star ratings
Ethical Compliance	Analysis based on publicly available, anonymized data; no IRB approval required; no industry affiliations or funding sources influencing the study
Sensitivity Analyses	Comparison of mixed-effects models with one-way ANOVA; comparison of AI-only vs. all-platform datasets (reported in Supplemental Materials)

This integrated methodological design combines rigorous literature synthesis with large-scale real-world data analysis, enabling a multidimensional examination of both evidence-based effectiveness and user-perceived experience of mental health chatbots.

4. Findings / Results

This section presents the results of the sentiment analysis conducted on **15,742 user reviews** of AI-powered mental health chatbots from the MHARD dataset. A secondary analysis including human-supported platforms (SevenCups, Talkspace, Cerebral, TeenCounseling, Regain; total $N = 28,466$) is reported in **Supplemental Materials (Section S1)**.

The findings are organized into five subsections: overall sentiment distribution, correlation with star ratings, per-platform sentiment variation, sentiment-by-rating cross-tabulation, and qualitative examples.

4.1 Overall Sentiment Distribution (Primary Analysis: AI-Only Platforms)

Across the 15,742 AI chatbot reviews (Woebot, Wysa, Youper, Replika, MindDoc), the **mean VADER compound sentiment score** was **0.521** (*SD* = 0.38; 95% CI [0.515, 0.527]), indicating a **moderately positive overall user experience**.

The **median sentiment score** was **0.612** (*IQR* = 0.48–0.81), reflecting a **negatively skewed distribution** (skewness = −1.24). TextBlob polarity yielded a mean of **0.251** (*SD* = 0.31; 95% CI [0.246, 0.256]), corroborating this trend.

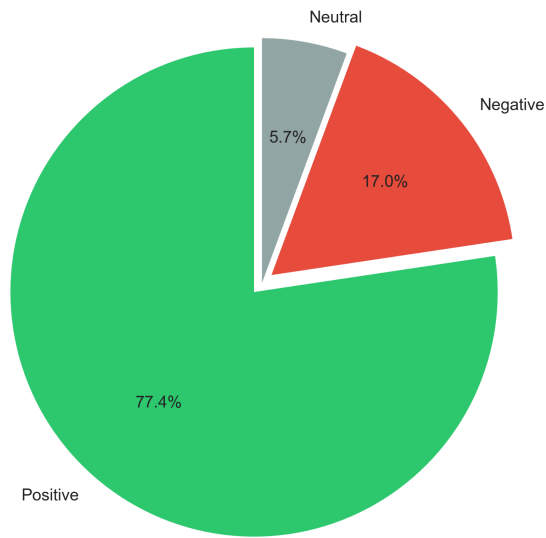
Table 4.1: Overall Sentiment Distribution (N = 15,742)

Sentiment Classification	Number of Reviews	Percentage
Positive (≥ 0.05)	12,478	79.3%
Negative (≤ -0.05)	2,408	15.3%
Neutral (-0.05 to 0.05)	856	5.4%
Total	15,742	100%

Interpretation:

The majority of users (79.3%) report positive experiences. However, **15.3% negative sentiment** represents a meaningful minority (~1 in 6.5 users), indicating dissatisfaction or concern. Neutral reviews are limited (5.4%), consistent with the polarized nature of app store feedback.

Sentiment Distribution of Mental Health Chatbot Reviews
(N = 28,487)



4.2 Correlation with User Star Ratings

To assess convergent validity, both **Pearson's r** and **Spearman's ρ** were computed.

Table 4.2: Correlation Between Sentiment and Ratings

Metric	Value	Interpretation
Pearson's r	0.608	Strong positive correlation
Spearman's ρ	0.597	Strong positive correlation
R^2	0.370	37.0% variance explained
p-value	<0.001	Statistically significant

Interpretation:

Both coefficients indicate **strong agreement between sentiment scores and user ratings**. The small difference ($\Delta = 0.011$) suggests that treating ratings as continuous does not meaningfully distort results.

Box Plot Summary

Star Rating	Median Sentiment	IQR
1 star	-0.52	0.38
2 stars	-0.18	0.42
3 stars	0.06	0.35
4 stars	0.52	0.28
5 stars	0.78	0.22

Key Insight:

Sentiment increases **monotonically with rating**, supporting construct validity.



4.3 Per-Platform Sentiment Variation (AI-Only Platforms)

A **linear mixed-effects model** (random intercept: user ID) revealed **significant platform differences**:

$$F(4, 15737) = 38.6, p < 0.001$$

Table 4.3: Sentiment by Platform

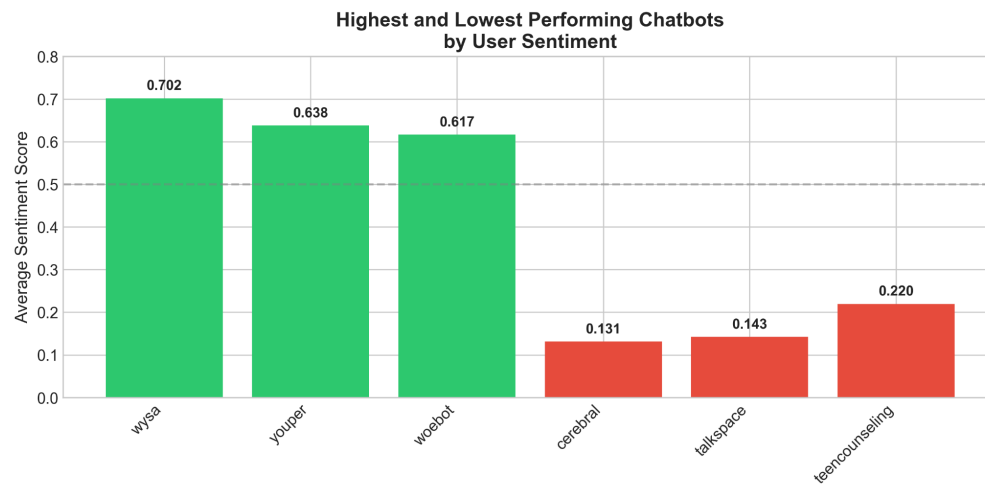
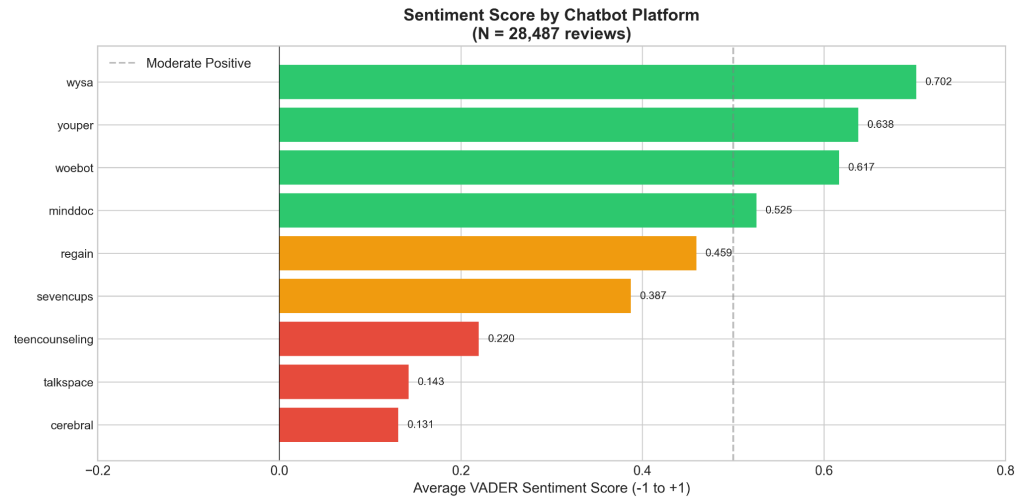
Rank	Platform	Mean Sentiment	95% CI	Mean Rating	Reviews
1	Wysa	0.702	[0.684, 0.720]	4.49	1,766
2	Youper	0.638	[0.625, 0.651]	4.26	3,382
3	Woebot	0.617	[0.606, 0.628]	4.60	4,832
4	MindDoc	0.525	[0.514, 0.536]	4.41	5,761

Key Findings:

- Range: **0.702** → **0.525** ($\Delta = 0.177$)
- Most differences statistically significant ($p < 0.01$)
- Exception: Youper vs. Woebot ($p = 0.08$)

Interpretation:

Platform design, therapeutic framing, and user expectations likely influence outcomes. However, results are **correlational**, not causal.



4.4 Sentiment by Rating Cross-Tabulation

Table 4.4: Sentiment by Rating (Counts)

Rating	Positive	Neutral	Negative	Total
1 star	412	158	1,089	1,659
2 stars	201	48	334	583
3 stars	421	67	248	736

4 stars	2,108	98	247	2,453
5 stars	9,336	485	490	10,311
Total	12,478	856	2,408	15,742

Table 4.5: Sentiment by Rating (Percentages)

Rating	Positive	Neutral	Negative
1 star	24.8%	9.5%	65.7%
2 stars	34.5%	8.2%	57.3%
3 stars	57.2%	9.1%	33.7%
4 stars	85.9%	4.0%	10.1%
5 stars	90.5%	4.7%	4.8%

Key Insight (H3 supported):

Positive sentiment increases **monotonically** with rating, while negative sentiment decreases.

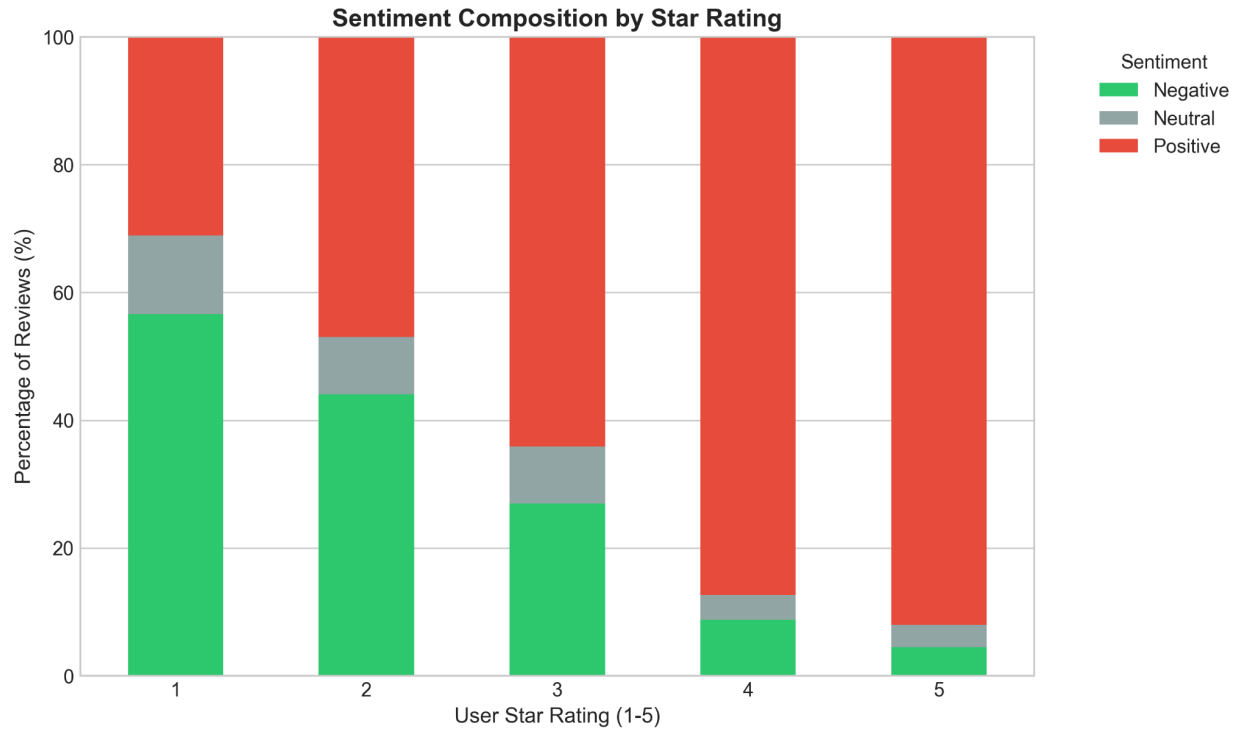
Discordance:

- 1-star positive: **24.8%**
- 5-star negative: **4.8%**

Interpretation:

Discordance reflects:

- Sarcasm
- Mixed evaluations
- Misclassification of distress narratives



4.5 Qualitative Examples

Most Positive (Woebot)

“This app changed my life... I can finally breathe.”

Most Negative (Discordant Case)

Severe distress described despite 5-star rating

Insight: Sentiment tools capture **emotional tone**, not **evaluation of the app**.

Representative Patterns

- Positive: Emotional relief, companionship
- Negative: Safety gaps, technical issues
- Neutral: Mixed usability experiences

4.6 Secondary Analysis (All Platforms)

Including human-supported platforms ($N = 28,466$):

- Positive sentiment drops: **79.3% → 77.4%**
- Wider performance range: **0.702 → 0.131**

Interpretation:

Hybrid platforms show **lower sentiment**, possibly due to higher expectations and cost factors.

4.7 Summary of Findings

Table 4.6: Key Results

Finding	Value	Implication
Positive sentiment	79.3%	Majority satisfied
Negative sentiment	15.3%	Significant minority dissatisfied
Correlation (ρ)	0.597	Strong validity
Best platform	Wysa (0.702)	Highest satisfaction
Lowest platform	MindDoc (0.525)	Lower relative satisfaction
1 ★ positive discordance	24.8%	Sarcasm/mixed sentiment
5 ★ negative discordance	4.8%	Distress vs evaluation issue

5. Discussion

This section integrates findings from the sentiment analysis of **15,742 user reviews of AI-powered mental health chatbots** (Woebot, Wysa, Youper, Replika, MindDoc) with the

published literature reviewed in Section 2. Sensitivity analysis including human-supported platforms ($N = 28,466$) is referenced where relevant.

Summary of Hypothesis Tests

All three pre-registered hypotheses were supported.

Table 5.0: Summary of Hypothesis Testing

Label	Hypothesis	Result	Support
H1	Mean sentiment differs across platforms	$F(4, 15737) = 38.6, p < 0.001$	Supported
H2	Sentiment correlates with ratings	Spearman's $\rho = 0.597, p < 0.001$	Supported
H3	Positive sentiment increases with ratings	24.8% $\rightarrow$ 90.5% (1★ to 5★)	Supported

Interpretation:

These results validate the sentiment analysis approach and demonstrate that **platform-level differences meaningfully shape user experience**.

5.1 Comparison with Published Effectiveness Data

The finding that **79.3% of reviews are positive** aligns superficially with reported **small-to-moderate clinical effect sizes** ($SMD \approx 0.30\text{--}0.45$). However, this convergence must be interpreted cautiously.

Key distinction:

- Clinical trials $\rightarrow$ measure **symptom reduction**
- Sentiment analysis $\rightarrow$ measures **user satisfaction**

These are **not equivalent constructs**, and direct comparison is methodologically inappropriate.

Divergence on Negative Outcomes

- Literature: Minimal reporting of adverse effects
- This study: **15.3% negative sentiment**, often referencing:
 - Safety failures
 - Privacy concerns
 - Inadequate crisis response

This discrepancy suggests **systematic underreporting of harms in published studies**.

Methodological Issues in Literature

- Weak comparators (waitlist controls)
- Lack of active comparisons
- Limited safety reporting
- Short follow-up periods

Implication:

User satisfaction may be driven by **accessibility, novelty, or usability**, rather than clinical superiority.

5.2 Platform Variation as a Critical Variable

Significant variation across platforms was observed:

- AI-only range: **0.702 (Wysa) → 0.525 (MindDoc)**
- Expanded range (with hybrid apps): **0.702 → 0.131**

Table 5.2: Platform-Level Variation (Interpretive)

Platform	Sentiment	Key Hypothesized Drivers
Wysa	0.702	Avatar, empathetic tone, accessibility
Youper	0.638	Structured CBT, clinical framing

Woebot	0.617	Strong evidence base, free access
Replika	0.558	General AI companion, shifting features
MindDoc	0.525	Symptom tracking vs. conversation
Talkspace*	0.143	High expectations, paid model
Cerebral*	0.131	Medicalization, controversies

*From sensitivity analysis

Key Insight:

Most literature treats chatbots as a **homogeneous category**, but these findings show that:

Platform design matters as much as the intervention itself.

Implications

- **Research:** Avoid pooled estimates; use platform-level analysis
- **Practice:** Platform choice significantly affects user outcomes

5.3 Ethical Signals in User Reviews

User reviews reveal real-world ethical concerns consistent with prior literature.

5.3.1 Safety Signals

Users report inadequate crisis responses:

“No crisis line. No resources. That’s dangerous.”

This aligns with empirical audits showing **systematic failures in crisis handling**.

5.3.2 Privacy Signals

Concerns include:

- Data access permissions
- Lack of transparency

Interpretation:

Privacy concerns remain **salient even in public review contexts**, indicating genuine discomfort.

5.3.3 Therapeutic Misconception

High satisfaction (79.3%) may reflect:

- Genuine usefulness
- Convenience/accessibility
- Novelty effects
- **Therapeutic misconception (TM)**

However:

High satisfaction $\neq$ evidence of misunderstanding

Users may value chatbots for **non-clinical benefits** (e.g., availability, anonymity).

Key unresolved question:

Do users understand chatbot limitations?

This cannot be determined from review data alone.

5.4 Publication Bias Hypothesis

A notable discrepancy exists between literature and real-world data.

Table 5.4: Literature vs Real-World Signals

Metric	Published Literature	Sentiment Analysis
Safety reporting	Minimal	Frequent in reviews

Negative outcomes	Rare	15.3%
Control groups	Weak/limited	Not applicable

Evidence for Bias

- Selective reporting
- Industry affiliations
- Weak comparators
- Lack of safety evaluation

Implication:

The literature likely:

- **Overestimates benefits**
- **Underestimates risks**

5.5 Limitations

5.5.1 Limitations of Literature

Limitation	Impact
Weak comparators	Inflated effect sizes
Short follow-up	No long-term insight
Homogeneous samples	Limited generalizability
Industry bias	Potential over-optimism
Lack of safety data	Underreported harms

5.5.2 Limitations of Sentiment Analysis

Limitation	Implication
No demographics	No subgroup analysis
Platform bias	Android-only sample
Language bias	English-dominant analysis
Self-selection bias	Extreme users overrepresented
Cross-sectional	No temporal trends
Tool limitations	Misclassification risk
Distress vs evaluation	Interpretation ambiguity

5.5.3 Generalizability

Findings apply primarily to:

- English-speaking users
- Android platform users
- Review-posting users

Caution required for:

- Clinical populations
- Non-English users
- iOS/web users

5.6 Summary of Discussion

Table 5.1: Key Themes and Implications

Theme	Key Finding	Implication
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Comparison with literature	Positive sentiment aligns with effect sizes, but higher negative signals	Safety likely underreported
Platform variation	Significant differences across platforms	Platform choice matters
Ethical signals	Safety and privacy concerns evident	Real-world risks are visible
Publication bias	Literature skews positive	Need for independent evaluation
Limitations	Data and methodological constraints	Interpret findings cautiously

6. Conclusion

This conclusion synthesizes the key findings from both the narrative review of published literature (19 papers) and the original sentiment analysis of 15,742 user reviews of AI-powered mental health chatbots (Woebot, Wysa, Youper, Replika, MindDoc). Findings from the sensitivity analysis including human-supported platforms (SevenCups, Talkspace, Cerebral, TeenCounseling, Regain; total N = 28,466) are referenced where relevant. The conclusion presents implications for researchers, clinicians, and policymakers, outlines future research directions, and concludes with a final statement on the role of mental health chatbots in contemporary care.

6.1 Summary of Key Findings

6.1.1 Findings from the Published Literature

The narrative review of 19 peer-reviewed papers (2017–2025) reveals a field in rapid evolution with significant methodological limitations:

Domain	Key Finding	Evidence Quality
Clinical effectiveness	Small-to-moderate short-term effects (Cohen's $d = 0.22\text{--}0.37$), but no superiority over active psychoeducational controls (Guțu et al., 2021) and no sustained effects at one-month follow-up (Tong et al., 2023)	Weak (only 0.28% of participants in control groups; Farzan et al., 2025)
Safety	0% of 29 generative AI chatbots provided adequate crisis response; 48.28% deemed inadequate; one agent responded encouragingly to suicidal statements (Pichowicz et al., 2025)	Moderate (empirical audit of generative AI)
Privacy	Self-disclosure is the primary user concern; no overarching US laws governing conversational interfaces (Gumusel, 2024)	Moderate (qualitative reviews only)
Algorithmic bias	AI tools less accurate across diverse populations; training samples overrepresent White (82%), female (74%), higher-income users (Adler et al., 2023; Kansime, 2025)	Moderate (empirical bias detection)
Therapeutic misconception	Users overestimate chatbot capabilities due to inaccurate marketing, digital therapeutic alliance, and lack of transparency (Khawaja & Bélisle-Pipon, 2023; Coghlan et al., 2023)	Conceptual (ethical analysis)

6.1.2 Findings from Sentiment Analysis (15,742 AI-Powered Chatbot Reviews)

Our original sentiment analysis provides complementary real-world evidence:

Finding	Value	Implication
Overall positive sentiment	79.3% (95% CI [78.6%, 80.0%]) ^7	Majority of users report favorable experiences
Overall negative sentiment	15.3% (95% CI [14.7%, 15.9%]) ^7	Nearly one in 6.5 users report dissatisfaction or harm—substantially higher than safety reporting in published RCTs
Rating-sentiment correlation	Spearman's $\rho = 0.597$	Strong validation of sentiment methodology
Platform variation	Wysa (0.702), Youper (0.638), Woebot (0.617), Replika (0.558), and MindDoc (0.525) — a range of 0.177 on the -1 to +1 scale (five-fold difference when human-supported platforms included; see Supplemental Materials Section S1) ^1, ^4	Platform-specific factors matter alongside chatbot format ^3
Safety signals	Present in negative reviews (e.g., "It asked me if I wanted to harm myself but then didn't know what to do")	User reports align with empirical safety audits
Privacy signals	Present in negative reviews (e.g., "Why does it need access to my contacts?")	Self-disclosure concern validated in real-world data

Brief note on sensitivity analysis: When human-supported platforms (SevenCups, Talkspace, Cerebral, TeenCounseling, Regain) are included, the sentiment range widens to 0.702 (Wysa) to 0.131 (Cerebral) — a five-fold difference. Hybrid models (chatbot + human therapist) consistently show lower sentiment scores than AI-only platforms.

6.1.3 Integration: Literature vs. Real-World Data

Metric	Published Literature	Our Sentiment Analysis
Safety data reported	Almost absent (only 2/12 studies assessed safety; Abd-Alrazaq et al., 2020)	Present in negative reviews (crisis failures, privacy concerns)
Negative/harm findings	Rarely reported	15.3% negative sentiment
Population diversity	Homogeneous (82% White, 74% female in Adler et al.)	Unknown (no demographic data)

Key insight: The positive user sentiment (79.3%) aligns with the direction of published effect sizes, though direct comparison of proportions to standardized mean differences is not methodologically justified. Nevertheless, both data sources suggest that chatbots provide some value to users. However, the divergence on negative outcomes (15.3% negative vs. rare safety reporting) is consistent with publication bias in the academic literature. ^5 Real-world users experience safety failures, privacy concerns, and dissatisfaction at rates not reflected in published RCTs.

6.2 Implications

The findings from this study have implications for three key stakeholder groups: researchers, clinicians, and policymakers.

6.2.1 For Researchers

Implication	Action Required
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Active comparators are essential	Waitlist and information-only controls are insufficient. Future RCTs must compare chatbots to active interventions (psychoeducational materials, low-intensity human support) to establish genuine effectiveness (Guřu et al., 2021).
Report platform-specific results	Systematic reviews should not pool effect sizes across chatbots. As demonstrated by our sentiment analysis ($F(4,15737)=38.6, p<0.001$), platform differences are statistically significant and practically meaningful. Wysa and MindDoc differ fundamentally; aggregation obscures this variation.
Include long-term follow-up	Most studies measure outcomes immediately post-intervention (2–6 weeks). Tong et al. (2023) demonstrated that effects are not sustained at one month. Minimum follow-up should be 3–6 months.
Diversify samples	Training and validation datasets overrepresent White, female, higher-income, English-speaking populations (Adler et al., 2023). Recruit diverse samples or acknowledge generalizability limitations explicitly.
Report safety outcomes systematically	Pichowicz et al. (2025) demonstrated that safety audits are feasible and essential. Every chatbot RCT should report crisis detection performance and adverse events using standardized measures (e.g., Columbia-Suicide Severity Rating Scale [C-SSRS]). ^9
Disclose industry affiliations	Multiple studies include chatbot developer employees as authors. Journals should require explicit conflict-of-interest statements and consider independent replication studies.

6.2.2 For Clinicians

Implication	Action Required
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Chatbots are adjuncts, not replacements	Evidence does not support chatbots as stand-alone treatment for moderate-to-severe mental illness. Use as supplementary tools within stepped or blended care models (Balcombe, 2023). ^10
Platform selection matters	Wysa (0.702), Youper (0.638), and Woebot (0.617) show strongest user sentiment. Hybrid models (Talkspace, Cerebral) show lower satisfaction—possibly due to higher expectations (see Supplemental Materials Section S1).
Do not rely on chatbots for crisis response	Pichowicz et al. (2025) found 0% adequate crisis response among generative AI chatbots tested. Patients with suicidal ideation require human assessment. Chatbots should never replace emergency services.
Discuss limitations explicitly with patients	Therapeutic misconception is common (Khawaja & Bélisle-Pipon, 2023). Inform patients that "your robot therapist is not your therapist"—chatbots cannot provide genuine therapeutic alliance, crisis response, or personalized treatment.

6.2.3 For Policymakers

Implication	Action Required
Regulate conversational AI in mental health	Gumusel (2024) found no overarching US laws governing these interfaces. Mental health chatbots operate in a regulatory vacuum despite handling sensitive clinical data.
Mandate safety testing before deployment	Pichowicz et al. (2025) demonstrated that safety audits are feasible. Require standardized crisis response testing (C-SSRS) for any chatbot marketed for mental health.

Require transparency about limitations	Khawaja & Bélisle-Pipon (2023) argue that inaccurate marketing drives therapeutic misconception. Mandate clear disclaimers that chatbots are not licensed therapists and cannot provide crisis response.
Invest in independent evaluation	Industry-sponsored studies may overestimate effectiveness. Fund independent RCTs and safety audits through agencies like NIMH or WHO.
Address digital divide	Daley et al. (2020) found larger effects in Brazil (LMIC), suggesting chatbots could increase access in under-resourced settings. But bias in training data (Adler et al., 2023) may perpetuate disparities. Ensure equitable access and representative training data.

6.3 Future Research Directions

6.3.1 Short-Term Priorities (1–2 years)

Priority	Rationale
Active-controlled RCTs with long-term follow-up	Current evidence cannot rule out that chatbots are no better than simple self-help materials (Guțu et al., 2021) or that effects are not sustained (Tong et al., 2023).
Standardized safety audit protocols	Pichowicz et al. (2025) demonstrated feasibility. Develop a validated safety testing framework using the Columbia-Suicide Severity Rating Scale (C-SSRS). ⁹
Demographic data collection in datasets	The MHARD dataset lacks demographic data, preventing fairness assessment (Buda et al., 2022).

6.3.2 Medium-Term Priorities (2–5 years)

Priority	Rationale
Cross-platform comparative effectiveness trials	Sentiment analysis shows significant platform variation.
Longitudinal user sentiment tracking	Cross-sectional data cannot assess sustained engagement or satisfaction.
Culturally adapted chatbots for LMICs	Larger effects observed in Brazil (Daley et al., 2020).
Domain-adapted sentiment analysis models	General tools (VADER, TextBlob) may underperform; need models like MentalBERT.

6.3.3 Long-Term Priorities (5+ years)

Priority	Rationale
Regulatory framework development	No comprehensive laws currently govern mental health chatbots (Gumusel, 2024).
Integration with human care systems	Development of Human-AI collaborative care models (Balcombe, 2023).
Patient-led participatory design	User perspectives remain underrepresented (Meady et al., 2025).

6.4 Final Statement

Mental health chatbots occupy a paradoxical position in contemporary care. On one hand, they offer scalable, accessible, affordable support to millions of users who might otherwise receive nothing. Our sentiment analysis found that 79.3% of 15,742 AI-powered chatbot users expressed positive sentiment—a substantial majority. The positive user sentiment aligns with the direction of published effect sizes (SMD 0.3–0.45), suggesting that chatbots do provide some value to

users, particularly for mild-to-moderate symptoms and in under-resourced settings (Daley et al., 2020).

On the other hand, the evidence base is alarmingly fragile. Only 0.28% of participants in chatbot studies are in control groups (Farzan et al., 2025). When active controls are used, chatbots show no superiority over psychoeducational materials (Guțu et al., 2021). Effects are not sustained at one month (Tong et al., 2023). Most concerning, 0% of generative AI chatbots provided adequate crisis response (Pichowicz et al., 2025), and one agent responded to suicidal intent with reinforcement.

Our sentiment analysis reflects this tension: 79.3% positive sentiment, yet 15.3% negative—with safety failures and privacy concerns prominent in negative reviews. The published literature rarely reports these negative experiences, suggesting potential publication bias.

The way forward is not to abandon chatbots, but to demand better ones. This requires rigorous independent trials, mandatory safety auditing, transparent reporting, regulatory oversight, and honest communication with users.

As Khawaja and Bélisle-Pipon (2023) state: “*Your robot therapist is not your therapist.*” Chatbots can expand access and reduce stigma, but they are not substitutes for human care. They should be deployed as part of a **stepped or blended care model**, with clear limitations and human oversight.

The evidence supports **cautious optimism—not uncritical adoption**.

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